

Stony Brook University
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Spring 2017

ASSIGNMENT 1

Due: Wednesday February 8, in class.

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you have used in problem XX to show your work.

1. Just prior to jury selection for O.J. Simpson's murder trial in 1995, a poll found that about 20% of the adult population believed Simpson was innocent (after much of the physical evidence in the case had been revealed to the public). *Ignore the fact that this 20% is an estimate based on a subsample from the population; for illustration, take it as the true percentage of people who thought Simpson was innocent prior to jury selection.* Assume that the 12 jurors were selected randomly and independently from the population (although this turned out not to be true).
 - (a) Find the probability that the jury had at least one member who believed in Simpson's innocence prior to jury selection. *Hint: Define the Binomial(12, .20) random variable X to be the number of jurors believing in Simpson's innocence.*
 - (b) Find the probability that the jury had at least two members who believed in Simpson's innocence *Hint: $\Pr(X \geq 2) = 1 - \Pr(X \leq 1)$ and $\Pr(X \leq 1) = \Pr(X = 0) + \Pr(X = 1)$.*
2. Suppose that a college student is taking three courses: a two-credit course, a three-credit course, and a four-credit course. The expected grade in the two-credit course is 3.5, while the expected grade in the three and four-credit course is 3.0. What is the expected overall grade point average for the semester? Remember that each course grade is weighted by its share of the total number of units.
3. Suppose in a survey of $N = 400$ students, 215 responded that they prefer small classes and 185 responded that they prefer large classes. Let p denote the fraction of all students who preferred small classes at the time of the survey, and $\hat{p}$ be the fraction of survey respondents who preferred small classes. (Hint: X is distributed as a Bernoulli random variable)
 - (a) Show that $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = p(1 - p)/N$.
 - (b) Use the survey result to estimate p , and calculate the standard error of your estimator. (Hint: Notice that this is the same as estimating the sample mean)
 - (c) What is the p-value for the test $H_0 : p = 0.5$ vs. $H_1 : p \neq 0.5$? Will you reject the null at the 5% significance level?
 - (d) Construct a 95% confidence interval for p .
4. Let Y_1, Y_2, Y_3 and Y_4 be independent, identically distributed random variables from a population with mean μ and variance σ^2 . Let $\bar{Y} = \frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4)$ denote the average of these four random variables.
 - (a) What are the expected value and variance of $\bar{Y}$ in terms of μ and σ^2

(b) Now, consider a different estimator of μ :

$$W = \frac{1}{8}Y_1 + \frac{1}{8}Y_2 + \frac{1}{4}Y_3 + \frac{1}{2}Y_4$$

Show that W is also an unbiased estimator of μ . Find the variance of W .

(c) Based on your answers to the previous questions, which estimator do you prefer $\bar{Y}$ or W ?

5. You are hired by the governor to study whether a tax on liquor has decreased average liquor consumption in your state. You are able to obtain, for a sample of individuals selected at random, the difference in liquor consumption (in ounces) for the years before and after the tax. For person i who is sampled randomly from the population, Y_i denotes the change in liquor consumption. Treat these as a random sample from a $\text{Normal}(\mu, \sigma^2)$ distribution.

- The null hypothesis is that there was no change in average liquor consumption. State this formally in terms of μ .
- The alternative is that there was a decline in liquor consumption; state this formally in terms of μ .
- Now, suppose your sample size is $n = 900$, and you obtain estimates $\bar{y} = -32.8$ and $s = 466.4$. Calculate the t statistic for testing H_0 against H_1 ; obtain the p -value for the test. (Because of the large sample size, just use the standard normal distribution.) Do you reject H_0 at the 5% level? At the 1% level?
- Would you say that the estimated fall in consumption is large in magnitude? comment on the practical versus statistical significance of this estimate.
- What has been implicitly assumed in your analysis about other determinants of liquor consumption over the two-year period in order to infer causality from the tax change to liquor consumption?

6. On Blackboard under the Problem Set 1 folder, you can find a zip file **CPS08.zip** which contains a database **CPS08.csv** for full-time, full-year workers age 25–34, with a high school diploma or B.A./B.S. as their highest degree. A detailed description is given in the pdf file **CPS08_Description**. In this exercise, you will investigate the relationship between a worker's age (*age*) and average hourly earnings (*AHE*).

(Hint: If you are looking for general methods to save plots in R, please look at the following webpage <http://www.stat.berkeley.edu/~s133/saving.html>)

- Load the dataset into R and call it **data_wage**. What command did you use? (**Do not use the "import dataset" command in Rstudio.**)
- How many observations are there in your sample? How many variables?
- Report the mean and the standard deviation for *AHE*. Also, create an histogram for *AHE*. Would you say that this variable is normally distributed?
- Make a scatter plot of *AHE* against *age* for the entire sample. What do you observe? Do average earnings increase with age or not? Explain.
- Repeat the previous question, but this time split the sample into males and females. Would you say that the relationship between age and *AHE* does not depend on gender? Explain.